12) a) Implement echo clientServer using TCP/UDP Sockets

Aim:
To implement echo clientServer using Tcp/Udp Sockets

procedure:

1) To write the Tcp code

```
import Socket
import threading
# Server
def start_Server():
    Server = Socket.Socket(Socket.AF_INET, Socket.
                           SOCK_STREAM)
    Server.bind(('127.0.0.1', 5000))
    Server.Listen().
    print("[Server] Listening on 127.0.0.1:5000
    Conn, addr = Server.accept()
    print(f"[Server] Connected {addr}")
    while True:
    data = Conn.recv(1024).decode()
    if not data or data.lower() == 'quit':
        printf("[Server] Connected closed by clie
        break.
```

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```
printf (f"[SERVER] Received: {data}")
conn.send(f"Server received {data}".encode())
Connection.close()
Server.close()
# Client # . . . .
def start_client():
    client = Server.Socket(socket.AF_INET, Socket.SOCK_
    STREM)
    client.connect(('127.0.0.1', 5000))
    print("[client] Connected to Server").
    while True:
        msg = input("Enter message (or 'quit to exit):"
        client.close()
# . . . Main - . . .
    if __name__ == "main__";
    Server_thread = thread.Thread(target = Start_Server,
    daemon = True)
    Server_thread.start()
#Start_client().
```

Result:

Therefore, the implement echo client server using TCP/UDP socket had successfully executed.

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output

Server listening on 127.0.0.1:5000 (client) connected to Server.

[Server] Connected by ('127.0.0.1', 53458)

HELLO

Hi.

[Server] connection closed by client

Result:

Therefore, the implement echo client / server TCP/UDP sockets had executed successfully.

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